LAU School of Engineering

ELE 537, Communication Systems
Final Exam, Fall 2018
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Problem 1: (10 points: (5×2))

a. Comment on one advantage and one disadvantage that follow from increasing the sampling frequency in Delta Modulation (DM) systems.

b. Consider a TDM system and denote by K the length of the synchronisation word. Comment on one advantage and one disadvantage that follow from increasing K.

c. Comment on one advantage and one disadvantage associated with Adaptive Delta Modulation (ADM).

d. Comment on one advantage and one disadvantage of DSB-SC-AM compared to conventional AM.

e. Comment on one advantage and one disadvantage of product detectors (used as AM demodulators) compared to envelope detectors.

Problem 2: (5 points: $(2,3)$)

a. A signal of bandwidth 1 MHz is transmitted over a channel of bandwidth 2 MHz. Will the communication system suffer from Inter-Symbol-Interference (ISI) in this case? Why?

b. A communication system has the equivalent impulse response given by $h_e(t) = \text{sinc}^2(200t)$. Find the maximum baud rate at which the system will satisfy the Nyquist first method for eliminating the ISI.

Problem 3: (5 points)
Delta Modulation (DM) is applied on the signal $w(t) = 10\cos(200\pi t)$. Find the value of the step-size that eliminates the slope overload noise while minimizing the granular noise if the DM system transmits at the rate of 10 kbits/s.

Problem 4: (10 points: $(7,3)$**)**

a. Design a TDM system that allows to accommodate the following sources:

- One analog source having a bandwidth of 100 Hz.
- One analog source having a bandwidth of 300 Hz.
- Three digital sources having a data rate of 235 bits/s.

where:

. Pulse stuffing is allowed and only two commutators can be used.

. The analog samples will be encoded into 3-bit PCM codewords.

. No synchronization bits are required.

Draw a block diagram for your design indicating the data rates at various points of your diagram.

b. The TDM signal is transmitted over a channel with overall raised cosine-rolloff Nuquist filter characteristics with $r = 0.5$. Find the absolute bandwidth required for the channel.

Problem 5: (10 points: $(2,2,3,3)$)

Different types of modulations are to be applied on the signal $m(t) = \cos(200\pi t)$ with $A_c = 10$. Find the normalized average power of the modulated signal in each of the following cases:

a. Double-Sideband Suppressed-Carrier (DSB-SC) Amplitude Modulation (AM).

b. Frequency Modulation (FM).

c. Amplitude Modulation (AM).

d. Upper Single-Sideband (USSB) Amplitude Modulation (AM).

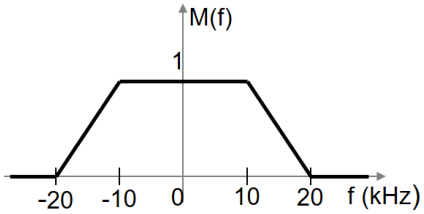
Problem 6: (10 points: $(3,3,4)$**)**
Amplitude Modulation (AM) is applied on the signal $m(t) = 2\sin(2\pi f_1 t)$ with $A_c = 1$ and the carrier frequency denoted by f_c.

a. The AM signal is applied to the input of a product detector. Write down the expression of the signal at the output of the detector. Is the detection successful? (*hint*: the signal at the output of the local oscillator is given by $A_o\cos(2\pi f_c t)$).

b. The AM signal is applied to the input of an envelope detector. Write down the expression of the signal at the output of the detector. Is the detection successful?

c. Provide a labeled sketch of the AM signal. On the same plot, sketch the signal that would appear at the output of an envelope detector.

Problem 7: (5 points)
Lower Single-Sideband (LSSB) Amplitude Modulation (AM) is applied on the signal whose spectrum is shown in the figure below with $f_c = 1$ MHz and $A_c = 100$:



Provide a labeled sketch of the spectrum of the LSSB signal (*hint*: find and plot $G(f)$ first).

Problem 8: (5 points)
A nonlinear system $x(t) \rightarrow y(t)$ is described by the relations $y(t) = x^3(t)$. Find the total harmonic distortion of the system (*hint*: let $x(t) = \cos(\omega_c t)$).

Problem 9: (5 points)
Find the normalized peak envelope power (PEP) of the RF signal $s(t) = 5\cos((\omega_c - \omega_a)t) + 5\cos((\omega_c + \omega_a)t)$ where $\omega_c \gg \omega_a$.

Problem 10: (6 points: $(2,2,2)$)
An FM radio is tuned to receive an FM broadcasting station of frequency 100 MHz. The radio is of the superheterodyne type with the local oscillator operating on the high side of the 100 MHz and using a 10 MHz intermediate frequency (IF).

a. Determine the frequency of the local oscillator.

b. Determine the frequency of the image response.

c. If the FM signal has a bandwidth of 200 kHz, find the requirements for the IF filter.

Problem 11: (9 points: $(3,3,3)$**)**
A modulated RF waveform is given by:

$$s(t) = 100\cos\left(2\pi 10^8 t + 4\cos(300\pi t)\right)$$

a. Determine the average power dissipated in a 50-Ω resistor.

Find the expression of the modulating signal in each of the following cases:

b. Phase Modulation (PM) is applied with a phase deviation constant of 100 rad/V.

c. Frequency Modulation (FM) is applied with a frequency deviation constant of 10^4 rad/(V.s).

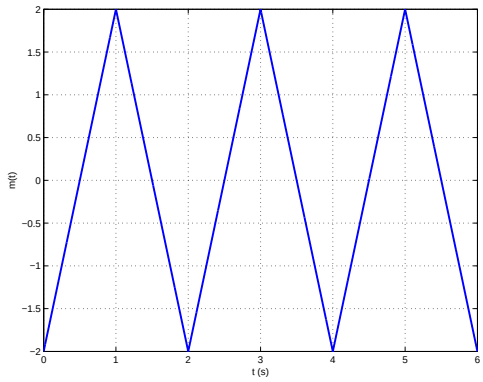
Problem 12: (5 points)
A phase-modulated (PM) waveform is given by:

$$s(t) = 10\cos\left(2\pi 10^6 t + \sin(2000\pi t)\right)$$

Provide a labeled sketch of the magnitude spectrum $|S(f)|$ for positive frequencies showing only the part of the spectrum included in the Carson's bandwidth.

Problem 13: (5 points)
Consider a diode envelope detector consisting of a diode, resistance R (Ohms) and capacitance C (Farads) that is used for AM demodulation. Assume that the carrier frequency is 1 MHz and that the modulating signal is bandlimited to 5 kHz. Find an appropriate range of values for RC.

Problem 14: (10 points: $(4,2,4)$**)**
A modulated signal is given by: $s(t) = \cos\left(2\pi f_c t + 5\int_{-\infty}^{t} m(\sigma)\mathrm{d}\sigma\right)$ where f_c is the carrier frequency and $m(t)$ is shown in the figure below:



a. Provide a labeled sketch of the instantaneous frequency.

b. Find the peak frequency deviation.

c. Which one of the below figures best describes the variations of $s(t)$? Justify your answer.

